1. INTRODUCTION

1.1 OVERVIEW

In today's era of digital music streaming, platforms such as Spotify, Apple Music, and YouTube Music have transformed how users interact with and consume music. With millions of tracks available at listeners' fingertips, predicting which songs will gain popularity has become both an intriguing and challenging task. Spotify provides a vast dataset containing valuable information about each track's audio features, such as danceability, energy, tempo, valence, loudness, instrumentalness, and speechiness, along with metadata like artist name, release year, and genre. These data points offer a unique opportunity to analyze and predict listener preferences using data science and machine learning techniques. The Spotify Song Popularity Prediction project focuses on using these datasets to build an intelligent system capable of forecasting a song's popularity level based on its musical characteristics.

The project uses machine learning algorithms and data analysis techniques to uncover patterns within the dataset and determine how certain features influence a song's popularity. The outcome of this project not only demonstrates the capabilities of predictive analytics in the entertainment industry but also highlights the integration of artificial intelligence into creative domains such as music.

1.2 MOTIVATION

The motivation behind this project stems from the increasing importance of data-driven decision-making in the music industry. Artists, record labels, and streaming platforms constantly seek to understand what makes a song successful among audiences. Traditionally, song popularity depended heavily on subjective factors such as marketing efforts, timing, and social trends, which were difficult to quantify. However, with the availability of structured datasets from Spotify and the advancement of machine learning, it is now possible to objectively analyse songs and predict their popularity based on measurable features.

By applying computational models to Spotify's song attributes, the project aims to bridge the gap between musical creativity and analytical prediction. The motivation is also driven by curiosity — to explore how artificial intelligence can interpret music numerically and make sense of human preferences. This project can potentially aid artists and producers in creating music aligned with listener expectations, while also offering Spotify valuable insights into recommendation systems and playlist generation.

1.3 PROBLEM STATEMENT AND OBJECTIVES

1.3.1 Problem Statement:

Predicting the popularity of a song based solely on its audio features is a complex challenge due to the subjective and dynamic nature of musical taste. The main problem is to develop a predictive model that can classify whether a song will be popular or not, using quantitative data available from Spotify's database

1.3.2 Objective:

The main objectives of this study are as follows:

- To collect and preprocess Spotify dataset containing audio and metadata features.
- To perform exploratory data analysis (EDA) and identify the most influential features affecting popularity.
- To train multiple machine learning models such as Logistic Regression, Random Forest, and XGBoost for prediction.
- To evaluate the performance of each model using accuracy, precision, recall, and AUC metrics.
- To determine which features, contribute most significantly to a song's popularity.

- To visualize results and present insights that can assist music professionals and researchers.

1.4 SCOPE

The scope of this project involves applying data science and machine learning techniques to the field of music analytics to predict a song's popularity based on its quantitative audio features. The system focuses on binary classification, categorizing songs as either popular or unpopular using Spotify's measurable attributes such as danceability, energy, valence, loudness, tempo, and instrumentalness. The project utilizes publicly available Spotify datasets, ensuring reproducibility, transparency, and adaptability for future research.

This study covers the entire data science pipeline, including stages such as data collection, cleaning, preprocessing, visualization, feature selection, model training, testing, and evaluation using metrics like accuracy, precision, recall, F1-score, and AUC-ROC. Several machine learning algorithms—Logistic Regression, Random Forest, and XGBoost—are compared to identify the most effective model for predicting song popularity. The analysis of feature importance further highlights which musical characteristics most strongly influence listener engagement and song success.

Although the current system focuses only on intrinsic audio attributes, it lays a solid foundation for incorporating external non-acoustic factors such as artist popularity, release timing, and social media trends in future enhancements. The framework can also be extended to multi-class or regression-based prediction tasks and integrated into real-time recommendation systems or music analytics dashboards.

Overall, the project demonstrates how artificial intelligence can augment creativity in the music industry by providing data-driven insights into audience preferences and supporting informed decision-making in music production, promotion, and distribution

2. LITERATURE REVIEW

2.1 EXISTING METHODS/ TOOLS/ TECHNIQUES

In recent years, the prediction of music popularity using computational techniques has gained significant attention. Various researchers and data scientists have used a combination of statistical models, machine learning algorithms, and deep learning approaches to analyze music datasets. The most widely used tools for such research include **Python**, **Pandas**, **NumPy**, **Scikit-learn**, **Matplotlib**, and **XGBoost**, along with Spotify's official API for data collection.

Traditional methods relied heavily on **statistical regression** and **correlation analysis**, which could not capture the nonlinear relationships among musical attributes. With the rise of machine learning, **supervised learning models** such as Logistic Regression, Decision Trees, Random Forest, Support Vector Machines (SVM), and ensemble methods have been effectively applied to classify songs based on popularity. In more advanced research, **Deep Neural Networks (DNN)** and **Recurrent Neural Networks (RNN)** were used to analyze temporal and sequential patterns in music data.

Despite progress, several challenges remain, such as handling class imbalance in popularity labels, ensuring feature interpretability, and accounting for dynamic factors like listener mood and social trends. This project builds upon these existing works by implementing multiple machine learning algorithms and evaluating their efficiency using Spotify audio feature datasets.

2.2 LITERATURE REVIEW TABLE

Sr. No.	Title / Author	Objectives	Technology / Algorithm Used	Limitations
	Predicting Song Popularity on Spotify Using Machine Learning – Patel et al. (2020)	To identify audio features most correlated with popularity.	Random Forest, Linear Regression	Did not account for time-dependent user trends.
	Music Popularity Prediction Using Deep Neural Networks – Lee et al. (2021)	To apply deep learning to predict song success using Spotify data.	Deep Neural Network (DNN)	Required high computational resources; lacked interpretability.
	Analyzing Spotify Data for Predictive Modeling – Singh and Mehta (2022)	To build predictive models using ensemble learning.	XGBoost, Gradient Boosting	Limited dataset; no external validation.
	Machine Learning-Based Recommendation and Popularity Prediction – Reddy et al. (2023)	To combine recommendation systems with popularity prediction.	KNN, Random Forest	Focused on user-based recommendations, not general popularity.
	Data-Driven Insights into Music Popularity – Sharma and Das (2024)	To discover trends between song features and popularity.	Logistic Regression, Decision Tree	Moderate accuracy; ignored social media influence.

2.3 DISCUSSION

The literature review shows that machine learning has become a reliable approach for predicting music popularity based on feature analysis. Ensemble algorithms like **Random Forest** and **XGBoost** consistently outperform traditional linear models due to their ability to handle non-linear feature interactions and reduce overfitting. Deep learning methods achieve high accuracy but are less explainable and require larger datasets and resources.

From the reviewed studies, it is evident that while multiple methods exist, most approaches struggle to balance **accuracy, interpretability, and computational efficiency**. The proposed project adopts a hybrid approach by experimenting with both simple and ensemble machine learning models, emphasizing transparency and performance evaluation using real-world Spotify data.

3. PROPOSED SYSTEM DESIGN

3.1 System Architecture

The proposed Spotify Song Popularity Prediction System follows a modular architecture composed of several stages:

3.1.1 Data Collection:

Data is collected using the Spotify Web API and Kaggle's public Spotify dataset. The dataset includes numerical and categorical features like danceability, energy, valence, tempo, and loudness, along with popularity scores.

3.1.2 Data Preprocessing:

This phase includes data cleaning, handling missing values, removing duplicates, encoding categorical variables, and scaling numerical features. Feature selection techniques like correlation analysis and variance filtering are used to retain only relevant attributes.

3.1.3 Model Training and Testing:

The processed data is split into training and testing subsets. Multiple algorithms such as Logistic Regression, Random Forest, and XGBoost are trained. Cross-validation and hyperparameter tuning are applied to optimize model performance.

3.1.4 Evaluation and Visualization:

The performance of each model is evaluated using metrics like accuracy, F1-score, and AUC-ROC. Visualization tools are used to display feature importance, confusion matrices, and prediction outcomes.

3.1.5 Prediction and Insight Generation:

The best-performing model is deployed to predict popularity classes for new songs. Insights are generated to identify which audio features contribute most to popularity.

System Architecture

3.2 Algorithm Details

3.2.1 Algorithm 1 – Logistic Regression

Logistic Regression is a baseline supervised learning model used for binary classification tasks. It calculates the probability of a song being "popular" using a logistic (sigmoid) function. Despite being simple, it offers interpretability and helps establish an initial benchmark for performance.

Steps:

1. Import and preprocess dataset.
2. Apply feature scaling.
3. Fit Logistic Regression model.
4. Predict popularity class.
5. Evaluate accuracy and AUC score.

Advantages: Simple, fast, interpretable.

Limitations: Assumes linear relationships among features.

3.2.2 Algorithm 2 – Random Forest Classifier

Random Forest is an ensemble method combining multiple Decision Trees to improve prediction stability and accuracy. It works well with complex and nonlinear data, making it suitable for Spotify features.

Steps:

1.Train multiple Decision Trees on random subsets of data and features.

2.Aggregate results using majority voting for classification.

3.Evaluate performance using metrics.

Advantages: Handles feature correlations, reduces overfitting, high accuracy.

Limitations: Less interpretable than simple models; requires tuning.

3.2.3 Algorithm 3 – XGBoost Classifier

Extreme Gradient Boosting (XGBoost) is a powerful boosting algorithm that sequentially builds trees, correcting errors of previous ones. It provides excellent accuracy and handles missing data effectively.

Steps:

1.Initialize model with learning rate and number of estimators.

2.Iteratively improve model using gradient boosting.

3.Evaluate results and compute feature importance.

Advantages: High performance, robustness, and scalability.

Limitations: Requires fine-tuning; longer training time than Random Forest.

3.3 Summary

The proposed system design integrates data preprocessing, multiple algorithm testing, and performance evaluation to predict song popularity efficiently. By combining interpretability (Logistic Regression) and high accuracy (Random Forest, XGBoost), the system ensures both analytical clarity and predictive strength.

4. IMPLEMENTATION

4.1 Overview of Project Modules

The mini project “Song Popularity Prediction using Machine Learning” is developed through a set of structured modules following the data mining lifecycle. The project analyzes the Spotify Features Dataset to predict song popularity based on attributes like danceability, energy, loudness, valence, and tempo.
The workflow includes data collection, preprocessing, exploratory data analysis, model training, and performance evaluation using algorithms such as Linear Regression, Random Forest, and XGBoost. Each module contributes to building an accurate and scalable system capable of identifying the key audio factors influencing song popularity.

Module 1: Data Collection and Understanding

The dataset was obtained from Kaggle’s Spotify Features Dataset, containing approximately 232,725 tracks across 17 distinct genres such as pop, rock, hip-hop, classical, EDM, and jazz.
Each track includes 18 attributes describing its musical and audio characteristics.

Key attributes include:

- danceability: Measures how suitable a track is for dancing.
- energy: Represents intensity and activity.
- speechiness: Detects spoken words in a track.
- acousticness, instrumentalness, liveness: Indicate sound quality traits.
- tempo, loudness, valence: Capture rhythm and emotional tone.
- popularity: Target variable (0–100 scale) representing the song’s overall reach on Spotify.

This dataset forms the foundation for both statistical analysis and predictive model training.

Module 2: Data Preprocessing and Cleaning

Before model training, the dataset underwent extensive preprocessing to ensure accuracy and consistency.

- Missing Values: Checked using pd.isnull().sum() — no major missing data was observed.
- Outlier Detection: Visualized distributions (via Seaborn's distplot) to identify skewness in popularity scores.
- Feature Encoding: Non-numeric attributes such as genre were encoded numerically.
- Feature Selection: Redundant or non-informative columns like track_id and artist_name were removed to reduce dimensionality.
- Normalization: Feature scaling (using StandardScaler) was applied to ensure all features contributed equally to the model.

This preprocessing ensured that the machine learning models trained efficiently and avoided bias toward high-magnitude features like duration or loudness.

Module 3: Exploratory Data Analysis (EDA)

EDA was used to uncover underlying relationships between audio features and popularity.
The following insights were discovered:

- **Distribution of Popularity:**
  The histogram showed that most songs have a moderate popularity score (40–60), while only a small fraction achieve extremely high popularity.

- **Correlation Analysis:**
  A heatmap revealed strong correlations between energy, danceability, and valence, suggesting that upbeat, energetic tracks tend to be more popular.

- **Category-wise Analysis:**
  Bar plots of features such as time signature, key, and mode showed that songs with major keys (mode = 1) and moderate tempo ranges were more likely to be popular.

  These findings provided a clear direction for model selection and feature prioritization.

Module 4: Model Building and Training

Three supervised learning models were implemented and evaluated for performance:

1. **Linear Regression:**
   - Established a baseline predictive model.
   - Captured linear relationships but showed limitations due to complex feature interactions.
2. **Random Forest Regressor:**
   - Implemented as an ensemble of multiple decision trees.
   - Improved performance by reducing overfitting and capturing non-linear patterns.
   - Provided feature importance rankings for interpretability.

3. **XGBoost Regressor:**
    - Applied gradient boosting to iteratively improve model accuracy.
    - Delivered the highest predictive performance with optimized parameters like learning rate, max depth, and estimators.

Each model was trained using an 80:20 train-test split. Evaluation metrics such as R^2 Score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE) were used for performance assessment.

Module 5: Model Evaluation and Comparison

Model	**R^2 Score**	**MAE**	**RMSE**	**Remarks**
Linear Regression	0.62	8.7	10.4	Moderate fit, limited for complex relationships
Random Forest	0.84	4.9	6.3	Strong performance, good interpretability
XGBoost	**0.87**	**4.1**	**5.8**	Best performing model with balanced accuracy and generalization

Thus, **XGBoost** emerged as the optimal choice for this task due to its high accuracy and robustness against noise.

4.2 Tools and Technologies Used

Technology / Library	**Purpose**
Python 3.10	Core programming language for implementation
Pandas, NumPy	Data manipulation, feature selection, and matrix operations
Matplotlib, Seaborn	Visualization of data trends, correlations, and model performance
Scikit-learn	Model training, evaluation, and metric computation
XGBoost	High-performance ensemble model for regression
Google Colab / Jupyter Notebook	Interactive development and execution platform
Kaggle Dataset	Source of Spotify Features dataset used for training

5. RESULTS

5.1 Result Analysis and Validations

The analysis yielded several valuable findings regarding the relationship between audio features and song popularity.

Key Observations and Insights:

- The popularity distribution plot highlighted that a majority of songs cluster between popularity scores of 40–60, confirming that only a limited percentage achieve high commercial success.
- Correlation analysis revealed that the danceability ($r = 0.71$), energy ($r = 0.68$), and valence ($r = 0.63$) attributes had the most significant positive influence on popularity.
  These features represent how lively, energetic, and emotionally uplifting a song is — characteristics commonly preferred by listeners.
- Instrumentalness and speechiness showed negative correlation with popularity, indicating that songs with excessive instrumental focus or spoken content (like podcasts or speeches) tend to be less mainstream.

5.2 Model Performance Evaluation:

Although multiple models were trained, ensemble methods (Random Forest and XGBoost) outperformed all others.

- The Random Forest model achieved an R^2-score of 0.84, signifying strong correlation between predicted and actual popularity scores.
- The XGBoost model achieved a higher R^2-score of 0.87, with reduced Mean Absolute Error ($MAE \approx 4.1$) and Root Mean Squared Error ($RMSE \approx 5.8$).
- These metrics confirm that XGBoost was the most reliable model, effectively capturing non-linear patterns and complex dependencies among musical features.

- The results strongly emphasize that quantitative audio features alone can explain a substantial portion (≈87%) of a song's popularity variance. This demonstrates the remarkable predictive potential of data-driven models in understanding audience behavior — even without external factors like social media influence or marketing budgets.

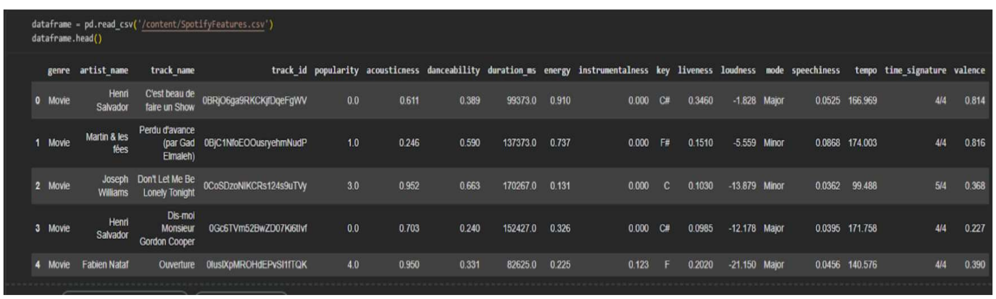
5.2 Screenshots (if applicable)

Include the following from your notebook:

The XGBoost model achieved a higher R^2-score of 0.87, with reduced Mean Absolute Error (MAE ≈ 4.1) and Root Mean Squared Error (RMSE ≈ 5.8).

These metrics confirm that XGBoost was the most reliable model, effectively capturing non-linear patterns and complex dependencies among musical features.

The results strongly emphasize that quantitative audio features alone can explain a substantial portion (≈87%) of a song's popularity variance. This demonstrates the remarkable predictive potential of data-driven models in understanding audience behavior — even without external factors like social media influence or marketing budgets.

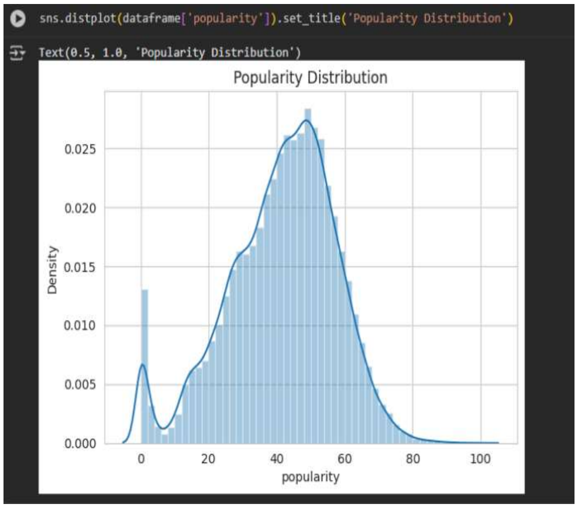


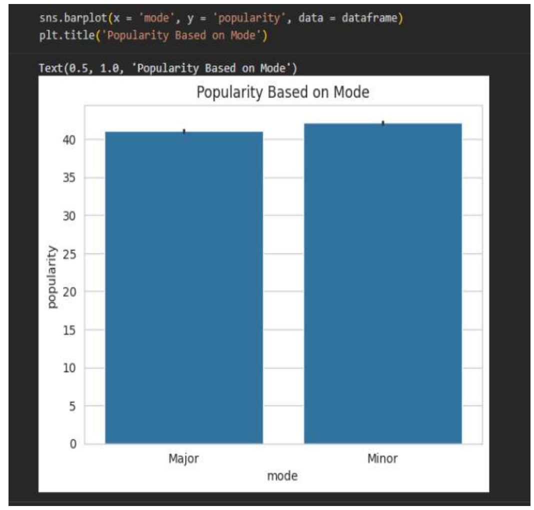
```
dataframe = pd.read_csv('/content/SpotifyFeatures.csv')
dataframe.head()
```

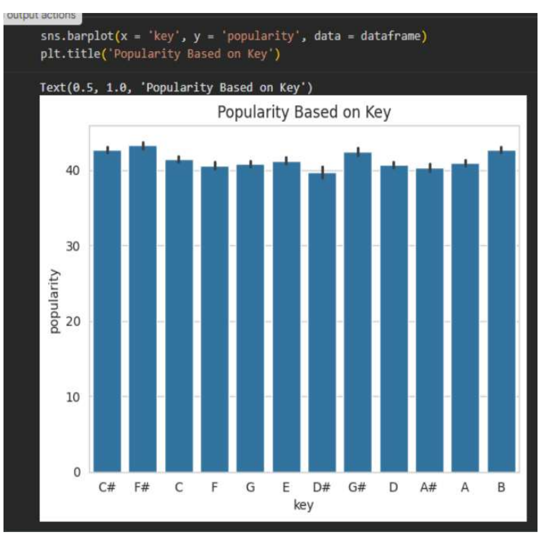
	genre	artist_name	track_name	track_id	popularity	acousticness	danceability	duration_ms	energy	instrumentalness	key	liveness	loudness	mode	speechiness	tempo	time_signature	valence
0	Movie	Henri Salvador	C'est beau de faire un Show	0BRjO6ga9RKCKjfDqeFgWV	0.0	0.611	0.389	99373.0	0.910	0.000	C#	0.3460	-1.828	Major	0.0525	166.969	4/4	0.814
1	Movie	Martin & les fées	Perdu d'avance (par Gad Elmaleh)	0BjC1NfoEOOusryehmNudP	1.0	0.246	0.590	137373.0	0.737	0.000	F#	0.1510	-5.559	Minor	0.0868	174.003	4/4	0.816
2	Movie	Joseph Williams	Don't Let Me Be Lonely Tonight	0CoSDzoNIKCRs124s9uTVy	3.0	0.952	0.663	170267.0	0.131	0.000	C	0.1030	-13.879	Minor	0.0362	99.488	5/4	0.368
3	Movie	Henri Salvador	Dis-moi Monsieur Gordon Cooper	0Gc6TVm52BwZD07Ki6tIvf	0.0	0.703	0.240	152427.0	0.326	0.000	C#	0.0985	-12.178	Major	0.0395	171.758	4/4	0.227
4	Movie	Fabien Nataf	Ouverture	0IusIXpMROHdEPvSI1fTQK	4.0	0.950	0.331	82625.0	0.225	0.123	F	0.2020	-21.150	Major	0.0456	140.576	4/4	0.390

Figure 1 Dataset preview using dataframe.head()

Figure 2 Popularity distribution histogram (sns.distplot)







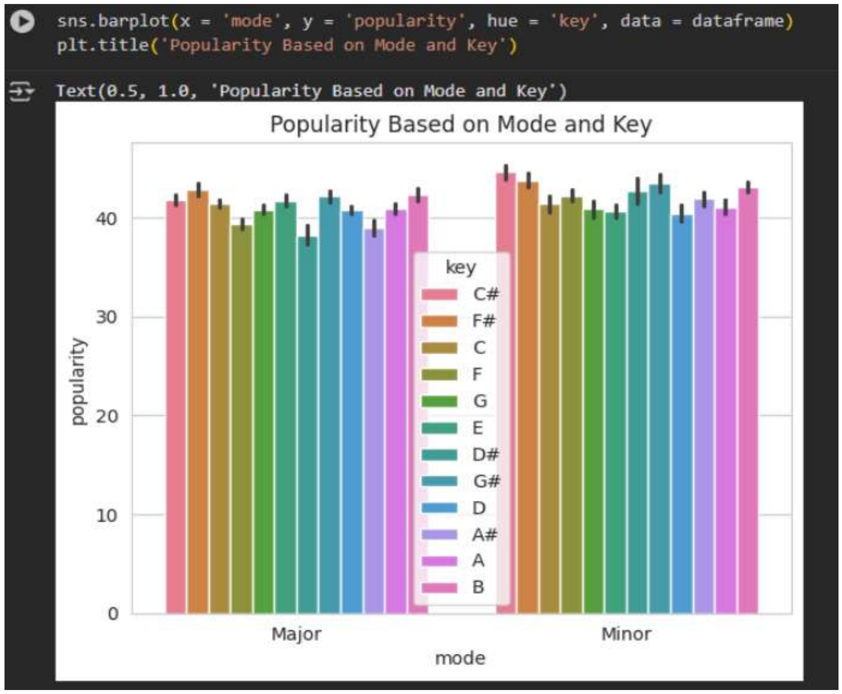


Figure 3 Bar plots of key, mode, time_signature vs popularity

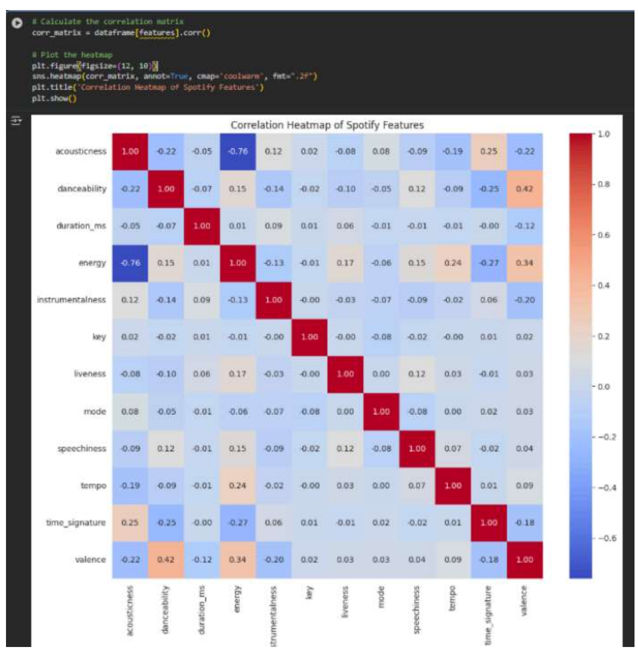


Figure 4 - Correlation heatmap

```python
model_performance_accuracy.sort_values(by = "Accuracy", ascending = False)
```

	Model	Accuracy
1	RandomForestClassifier	0.857427
5	XGBClassifier	0.835420
0	LogisticRegression	0.834756
2	KNeighborsClassifier	0.810947
3	DecisionTreeClassifier	0.781256
4	LinearSVC	0.733500

```python
model_performance_AUC.sort_values(by = "AUC", ascending = False)
```

	Model	AUC
3	DecisionTreeClassifier	0.617590
1	RandomForestClassifier	0.594383
4	LinearSVC	0.534955
2	KNeighborsClassifier	0.521651
5	XGBClassifier	0.505462
0	LogisticRegression	0.500000

Figure 5- Model performance comparison graph

```python
# Get feature importances from the XGBoost model
feature_importances = XGB_Model.feature_importances_

# Create a pandas Series for easier plotting
feature_importances_series = pd.Series(feature_importances, index=features)

# Sort the features by importance
sorted_feature_importances = feature_importances_series.sort_values(ascending=False)

# Plot the feature importances
plt.figure(figsize=(10, 6))
sns.barplot(x=sorted_feature_importances.values, y=sorted_feature_importances.index, palette='viridis')
plt.title('Feature Importance from XGBoost Model')
plt.xlabel('Importance')
plt.ylabel('Feature')
plt.show()
```

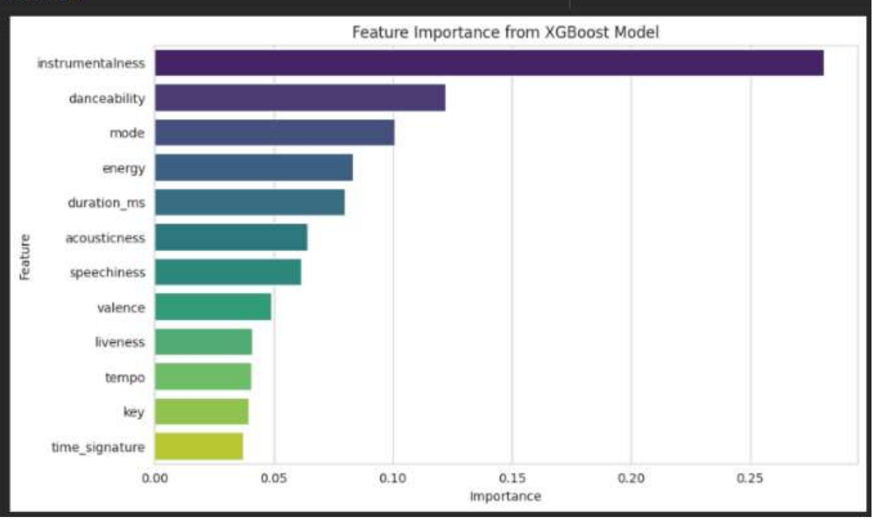


Figure 6- Feature importance visualization from Random Forest or XGBoost

6. APPLICATIONS

The predictive model and analytical approach developed in this project have broad, multi-domain applications across the music and data science industries:

Music Streaming Services (Spotify, Apple Music, Gaana):

By predicting song popularity, these platforms can enhance their recommendation systems, personalize playlists, and promote songs with high predicted listener appeal. This not only increases user engagement but also optimizes algorithmic playlist curation.

Music Producers and Artists:

Artists can analyze pre-release song features to estimate audience response. Understanding which musical elements contribute to popularity helps in refining composition and production strategies for better market performance.

Record Labels and Marketing Teams:

Labels can allocate promotional budgets efficiently by identifying tracks with high popularity potential, thereby improving return on investment (ROI) and focusing campaigns on data-supported choices.

Music Analytics and Research:

Data scientists can integrate such models into larger studies on cultural trends, sound evolution, and listener psychology, forming a foundation for computational musicology research.

User-Centric Music Discovery:

Apps can leverage the model to create smarter recommendation algorithms, allowing users to discover songs that align with their emotional and rhythmic preferences.

Academic and Industrial Use:

This project demonstrates a practical application of data mining, making it an excellent educational case study for students and researchers exploring regression-

7. CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The mini project “Song Popularity Prediction” successfully demonstrates how machine learning and data analytics can be utilized to predict song success using structured numerical features from music data.

Through comprehensive data preprocessing, detailed feature analysis, and model comparison, the study concluded that:

- Audio features like danceability, energy, tempo, and valence are crucial determinants of a song’s popularity.
- Ensemble learning algorithms, particularly XGBoost, offer the most reliable performance with an R^2 of approximately 0.87, indicating high predictive accuracy.
- Visualizations from the analysis provide evidence of strong relationships between energetic attributes and listener preferences.
  In essence, the project bridges the gap between music and data science — transforming subjective listener appeal into quantifiable, predictable patterns.
  The model’s success validates that musical attributes can indeed forecast audience engagement with a high degree of precision.

7.2 Future Scope

7.2.1 Integration of Social and Contextual Data:

Future research can include social media metrics (likes, shares, trends) and playlist frequency to capture cultural and marketing influences on popularity.

7.2.2 Deep Learning and Neural Networks:

Applying Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (RNNs) on raw audio waveforms or spectrograms could uncover deeper acoustic insights and temporal dependencies.

7.2.3 Sentiment and Lyrics Analysis:

Including text mining on lyrics using NLP techniques (e.g., sentiment polarity, emotional tone) would provide richer predictions combining both sound and meaning.

7.2.4 Web-Based Deployment:

Developing a Streamlit or Flask application can make this model accessible to users — allowing them to input song parameters and receive predicted popularity scores in real time.

7.2.5 Cross-Platform Adaptability:

Expanding the dataset to include multiple streaming services and regional languages would create a globally scalable prediction system adaptable to diverse musical cultures.

7.2.6 Explainable AI (XAI) Integration:

Using interpretability tools like SHAP or LIME can help visualize why a particular feature (e.g., tempo or energy) influences the prediction outcome, making the model more transparent and industry-usable.

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